TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	60.3 ± 3.40
Correlation Stability	78.0 ± 1.41
Distribution Consistency	78.0 ± 3.74
Entropy Preservation	81.0 ± 1.41
Mean Z Score Drift	0.6 ± 0.04
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	27.0 ± 4.55
Data Density	100.0 ± 0.00
Coverage Score	46.3 ± 7.36
Feature Accuracy	67.7 ± 1.70
Novelty Score	0.5 ± 0.03

TabuLLM: AI Benchmark Report

Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	3
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	180.193	+25.964	16.8%	Warn
MDVP:Fhi	197.105	205.529	+8.424	4.3%	OK
MDVP:Flo	116.325	143.016	+26.691	22.9%	Warn
MDVP:Jitter	0.006	0.004	-0.002	38.8%	Drift
MDVP:Jitter.1	0.006	0.004	-0.002	38.8%	Drift
MDVP:RAP	0.003	0.002	-0.001	44.6%	Drift
MDVP:PPQ	0.003	0.002	-0.001	42.8%	Drift
Jitter:DDP	0.010	0.005	-0.004	44.6%	Drift
MDVP:Shimmer	0.030	0.018	-0.012	40.4%	Drift
MDVP:Shimmer.1	0.030	0.018	-0.012	40.4%	Drift
Shimmer:APQ3	0.016	0.010	-0.006	39.0%	Drift
Shimmer:APQ5	0.018	0.010	-0.007	41.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	60.3 ± 3.40
Correlation Stability (*2)	78.0 ± 1.41
Distribution Consistency (*3)	78.0 ± 3.74
Entropy Preservation (*4)	81.0 ± 1.41
Mean Z Score Drift (*5)	0.6 ± 0.04
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	27.0 ± 4.55

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	46.3 ± 7.36
Feature Accuracy (*10)	67.7 ± 1.70
Novelty Score (*11)	0.5 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	N/A	0.9743589743589745	3.700743415417188e-17	
F1	0.980643539467069	N/A	0.979591836734694	-0.0010517027323749817	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.888888888888889	N/A	0.8803418803418804	-0.008547008547008553	
F1	0.915702947845805	N/A	0.9058096814566596	-0.009893266389145486	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9572649572649571	N/A	0.905982905982906	-0.05128205128205126	
F1	0.9685551485551486	N/A	0.9300847457627118	-0.038470402792436666	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9572649572649571	N/A	0.9572649572649573	3.700743415417188e-17	
F1	0.968042328042328	N/A	0.9662585034013605	-0.0017838246409674536	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	N/A	0.7521367521367521	-0.02564102564102566	
F1	0.8614267676767677	N/A	0.8128571428571428	-0.04856962481962487	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	N/A	0.7692307692307692	-0.11111111111111109	
F1	0.91232423490488	N/A	0.8132505175983437	-0.0990737173065364	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	N/A	0.7948717948717948	-0.05128205128205129	
F1	0.8918089248277927	N/A	0.8372234935163997	-0.05458543131139307 5	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	N/A	0.7008547008547009	-0.01709401709401710 6	
F1	0.749906191369606	N/A	0.7286324786324787	-0.02127371273712737 7	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7435897435897436	N/A	0.7692307692307692	0.02564102564102562 3	
F1	0.8329949937721568	N/A	0.8170250109697236	-0.01596998280243318 6	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	N/A	0.8461538461538461	-0.09401709401709402	
F1	0.9570703384745131	N/A	0.8805133400878081	-0.07655699838670489	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	240.524134059882	250.896009301906	230.269477401150	0.00276411858303	0.00276411858303	0.00155013904146	0.00154698977078	0.00464650710354	0.01385106762662	0.01385106762662
	78	13	14	1599	1599	25522	45234	9886	4042	4042
2	238.229045744290	249.729906861958	229.486856797931	0.00260896982134	0.00260896982134	0.00145625924102	0.00157129196422	0.00436142285701	0.01679852441932	0.01679852441932
	1	63	67	97014	97014	68047	34483	0023	8414	8414
3	122.372261762215	129.507509576710	114.204575628600	0.00420509642559	0.00420509642559	0.00126808884040	0.00158433080493	0.00381348627563	0.01667111628912	0.01667111628912
	47	9	47	6896	6896	11641	4756	67925	7576	7576
4	124.776209612526	135.393911821897	117.570099460292	0.00434995372549	0.00434995372549	0.00142625236291	0.00168083490860	0.00426943427052	0.02205263508144	0.02205263508144
	06	38	67	3841	3841	27491	84996	6278	18	18
5	116.869032189788	207.698419084095	99.5596251263536	0.00507001732164	0.00507001732164	0.00248674974729	0.00328742818635	0.00746024924187	0.01776394601340	0.01776394601340
	48	5	1	42265	42265	27814	0905	8343	957	957
6	238.518612007488	245.117108778603	229.877460198137	0.00284819503483	0.00284819503483	0.00160488690569	0.00154699237383	0.00481466071707	0.01276601590144	0.01276601590144
	74	9	13	2484	2484	11718	13403	3515	7852	7852
7	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
8	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
9	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
10	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	203.093800973333	211.533663969586	196.250297282687	0.00178196512040	0.00178196512040	0.00093901743979	0.00106098256020	0.00282508719898	0.00957606975918	0.00957606975918
	26	1	1	66828	66828	66585	33414	3293	6635	6635
2	174.625453280489	231.481152147345	94.0067518869681	0.00462726583082	0.00462726583082	0.00265362812252	0.00261172518648	0.00797056721156	0.04063007148068	0.04063007148068
	95	26	9	547	547	33833	1231	3124	538	538
3	218.554645308913	227.767677232020	203.528489386001	0.00269365308342	0.00269365308342	0.00154048448556	0.00147511626096	0.00462999027914	0.01563256824060	0.01563256824060
	65	28	58	40804	40804	24836	42579	72735	3315	3315
4	239.744593667625	255.398215974394	229.750023120518	0.00347632255428	0.00347632255428	0.00204683320896	0.00199841660448	0.00614049962688	0.01799037104817	0.01799037104817
	7	33	42	48155	48155	29395	147	8818	842	842
5	209.224279176788	253.382122032668	89.0637199083753	0.00568898438002	0.00568898438002	0.00334052103678	0.00295918750402	0.01003080950451	0.02744330588257	0.02744330588257
	8	1	6	7711	7711	64965	21685	2909	7654	7654
6	216.066311778532	225.246060832248	202.894895038964	0.00239430486377	0.00239430486377	0.00133772077863	0.00132568930185	0.00401698337924	0.01430134943185	0.01430134943185
	2	6	82	2544	2544	46523	56294	7505	2125	2125
7	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
8	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
9	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
10	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	227.880868492730 2	241.078022815718 4	110.589780876151 2	0.00262386598493 9893	0.00262386598493 9893	0.00147785620548 76688	0.00153119841742 96688	0.00443356861646 3006	0.01755303152555 2112	0.01755303152555 2112
2	198.470870412517 63	214.611200570842 87	192.998659493632 74	0.00215015670962 14322	0.00215015670962 14322	0.00115183761731 20716	0.00135311965961 60103	0.00345551285193 62147	0.01237210813840 9822	0.01237210813840 9822
3	117.507707389982 13	124.225658078816 76	107.049742291330 9	0.00417540045536 7381	0.00417540045536 7381	0.00182956106974 74868	0.00264354069388 6467	0.00548917415973 0404	0.01897806328874 882	0.01897806328874 882
4	202.750096006941 22	221.917130667865 73	92.3746801664842 4	0.00694742489233 16355	0.00694742489233 16355	0.00422514798483 011	0.00388347902230 51864	0.01267343546634 9943	0.02122569971395 524	0.02122569971395 524
5	116.416697707757 15	585.038196630932 7	91.4472793420706 9	0.00346933528394 3391	0.00346933528394 3391	0.00163518604571 23639	0.00188141291551 74232	0.00490555813713 7091	0.01425659695292 4108	0.01425659695292 4108
6	212.529244518936 08	250.891224909028 58	88.4078979662349	0.00479801716310 3832	0.00479801716310 3832	0.00283481029786 22996	0.00273956791851 2649	0.00849443089358 69	0.02241245456965 8978	0.02241245456965 8978
7	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
8	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
9	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
10	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645